Internship Title: 2025

Internship Program Name: Cyber Security

Your Name: Areesh Awan

Internship Lead Name: Faizyab



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H.12, Islamabad, Pakistan

## Contents

Internship Title: ..... 1

Internship Program Name: ..... 1

Your Name: ..... 1

Internship Lead Name: ..... 1

Date of Submission ..... 1

Executive Summary ..... 3

1. Introduction ..... 3
2. Task details and working ..... 3
3. Learning Outcomes ..... 3
4. Conclusion. ..... 3



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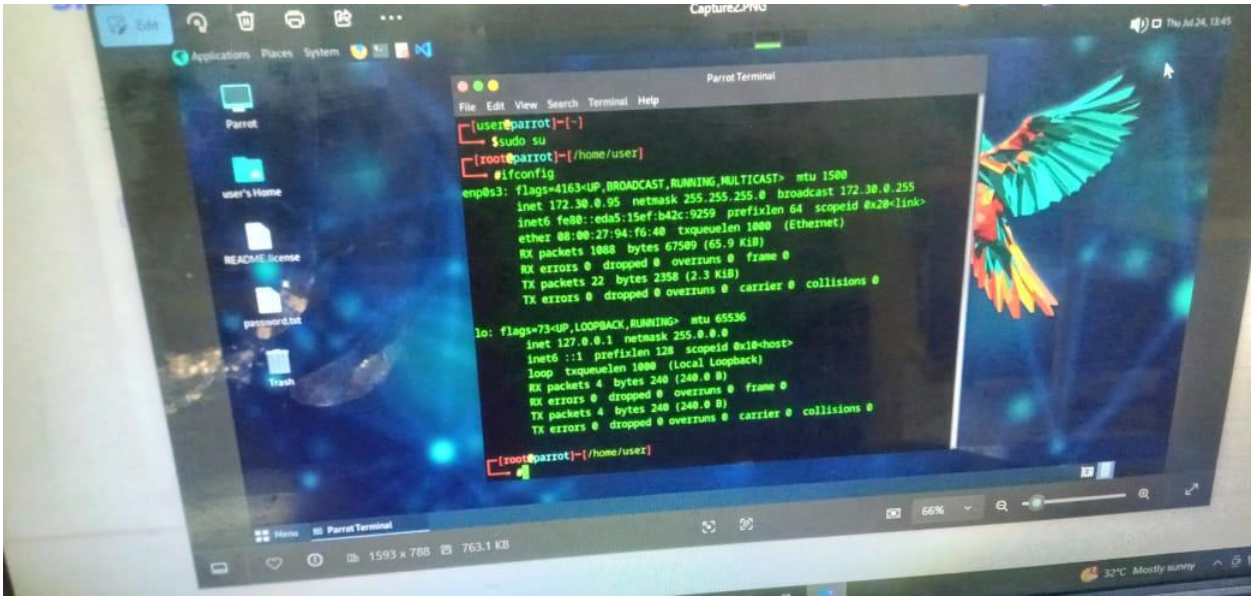
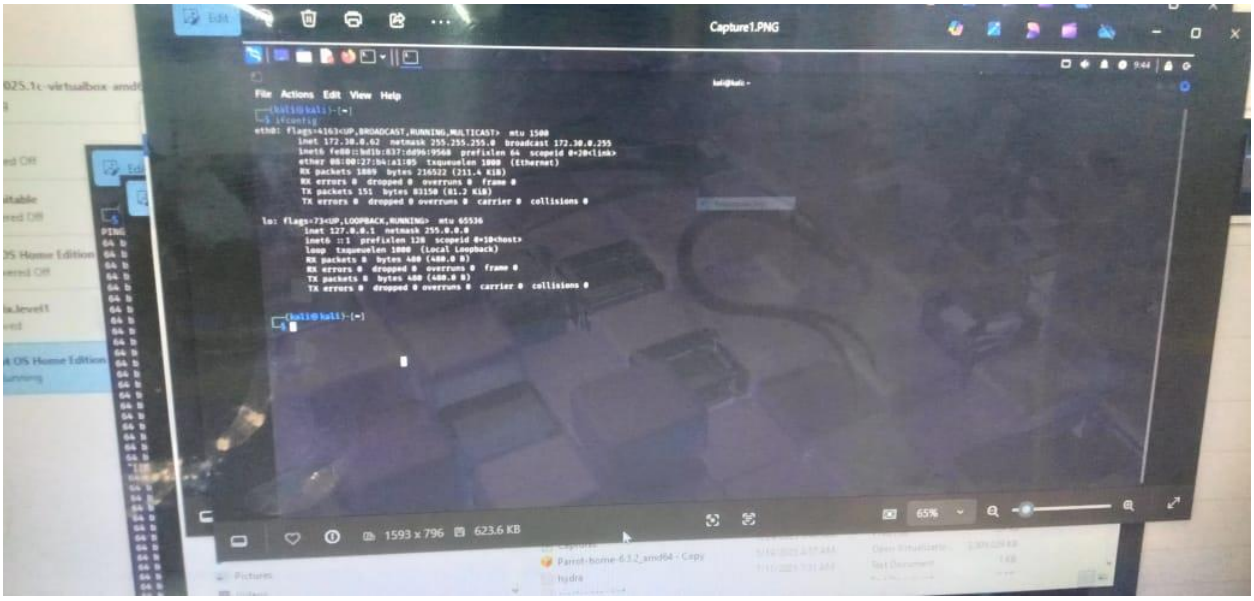
## Introduction

In this project, I used VirtualBox to connect two virtual machines – Kali Linux and Parrot OS – using different network modes: NAT, Bridged, and Internal Network. These modes help in understanding how virtual machines can communicate with each other and with the internet. This setup is useful for practicing ethical hacking, penetration testing, and secure networking in a safe environment.

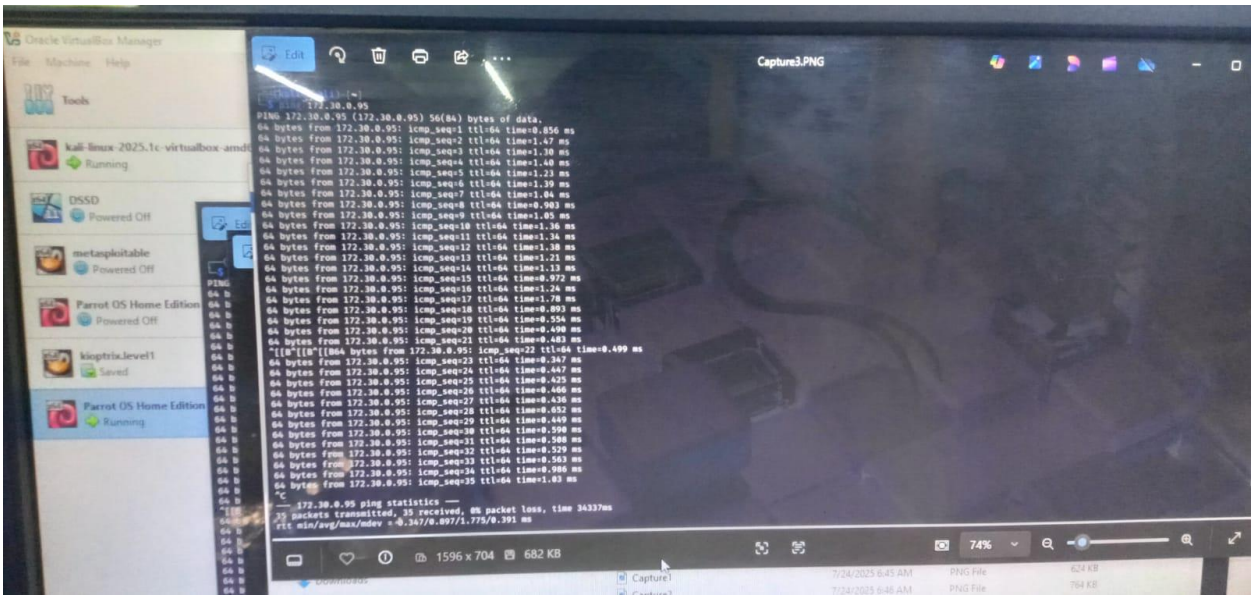
## Task Detail and Working





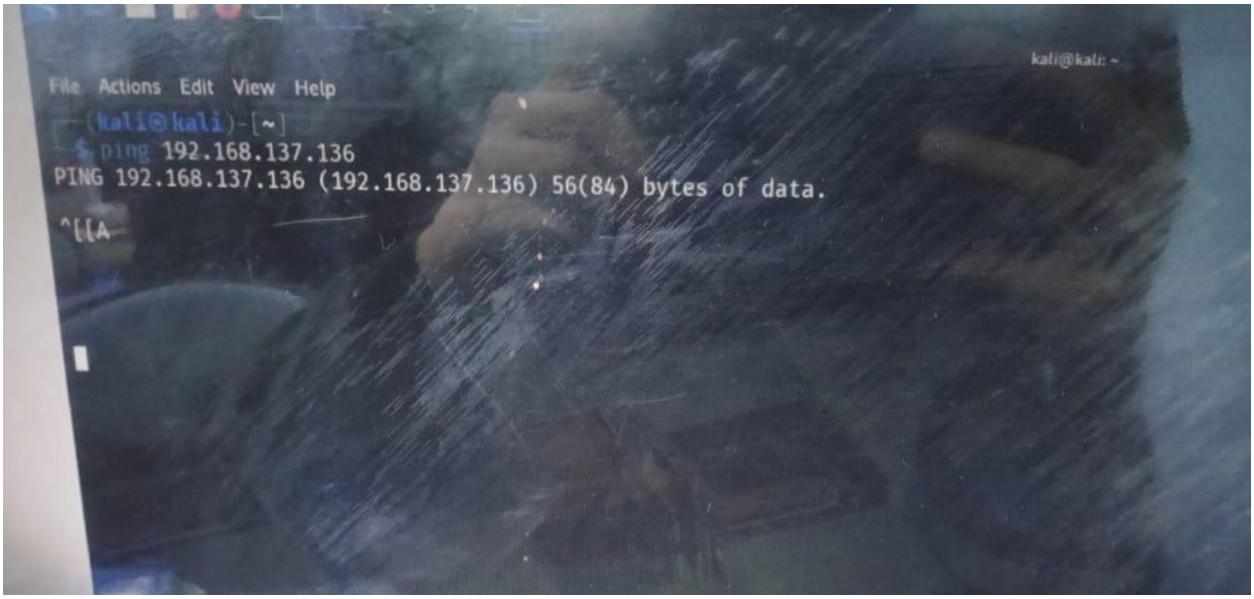

## Bridged Adapter

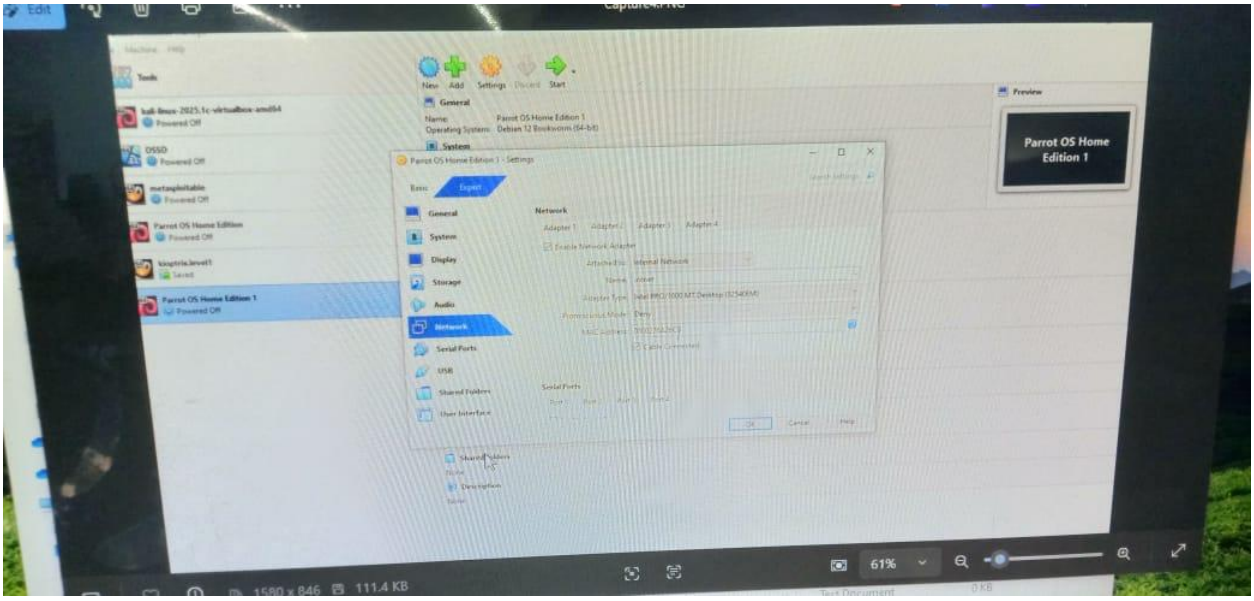
In VirtualBox, a Bridged Adapter is a network mode that allows a virtual machine to connect directly to the physical network as if it were a separate physical device. When this mode is enabled, the VM uses the host computer's network interface (such as Wi-Fi or Ethernet) to communicate with other devices on the same network. As a result, the VM receives its own IP address from the router or DHCP server, just like any other device on the network. This setup enables the virtual machine to interact with other computers, printers, and servers on the local network, and allows other devices to access the VM as well. The Bridged Adapter mode is ideal for scenarios where full network visibility is required, such as in server hosting, remote access testing, or network simulation environments.




## NAT

In VirtualBox, NAT (Network Address Translation**)** is a network mode that allows a virtual machine (VM) to access the internet through the host computer's network connection. The VM is placed behind a virtual router created by VirtualBox, which means it shares the host's IP address and routes its traffic through the host.

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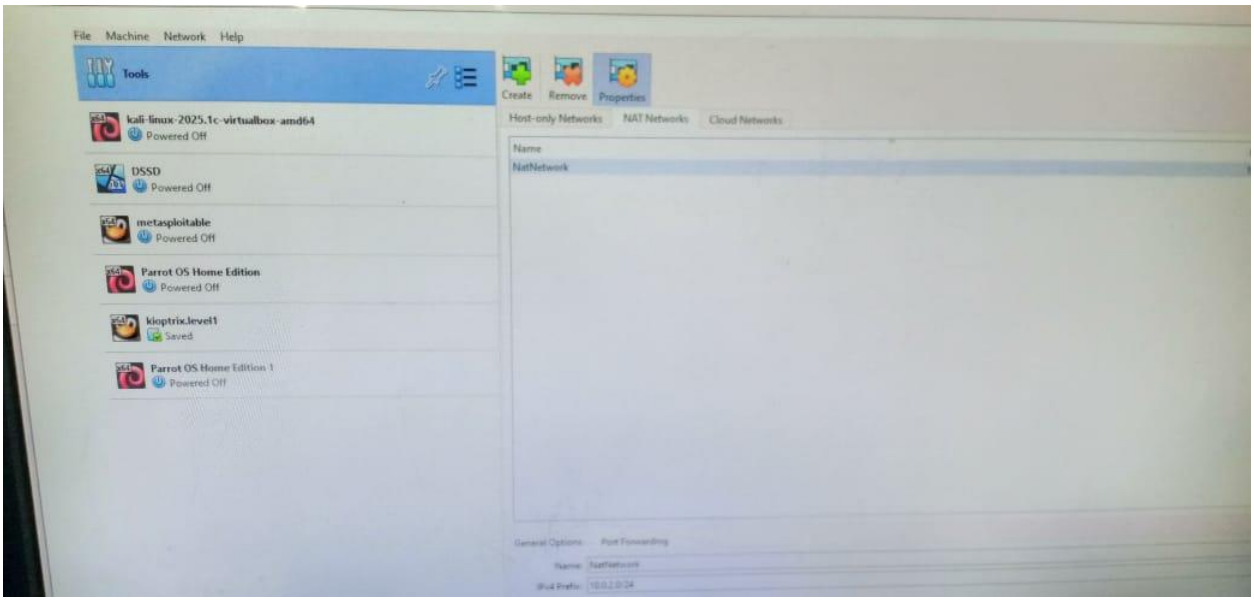
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DSSD
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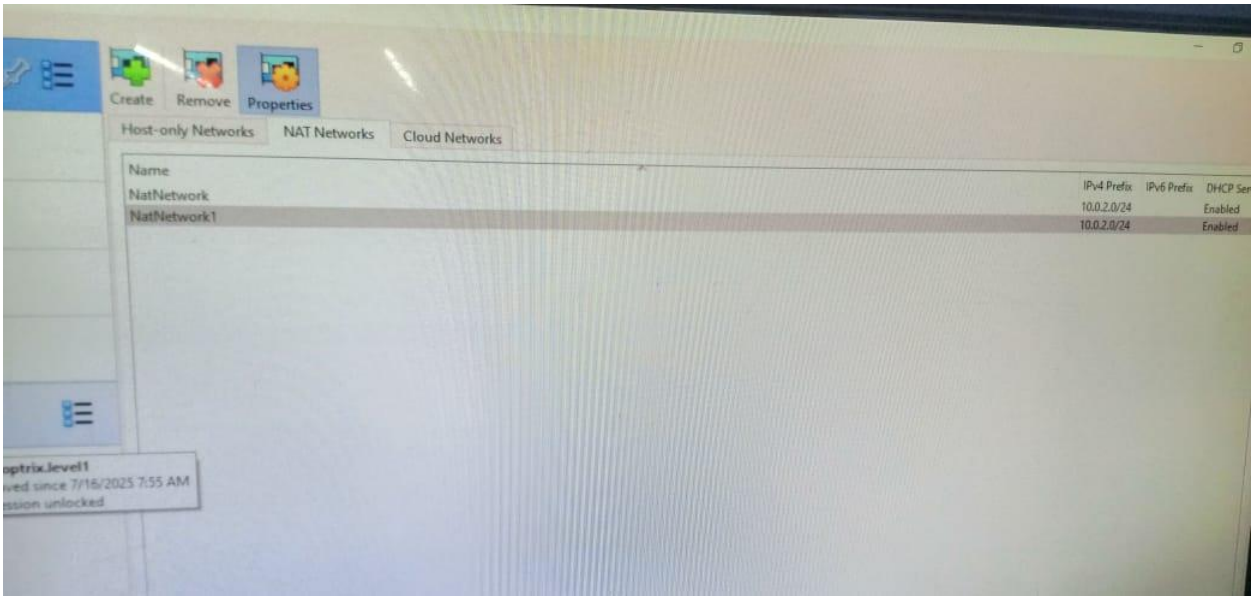
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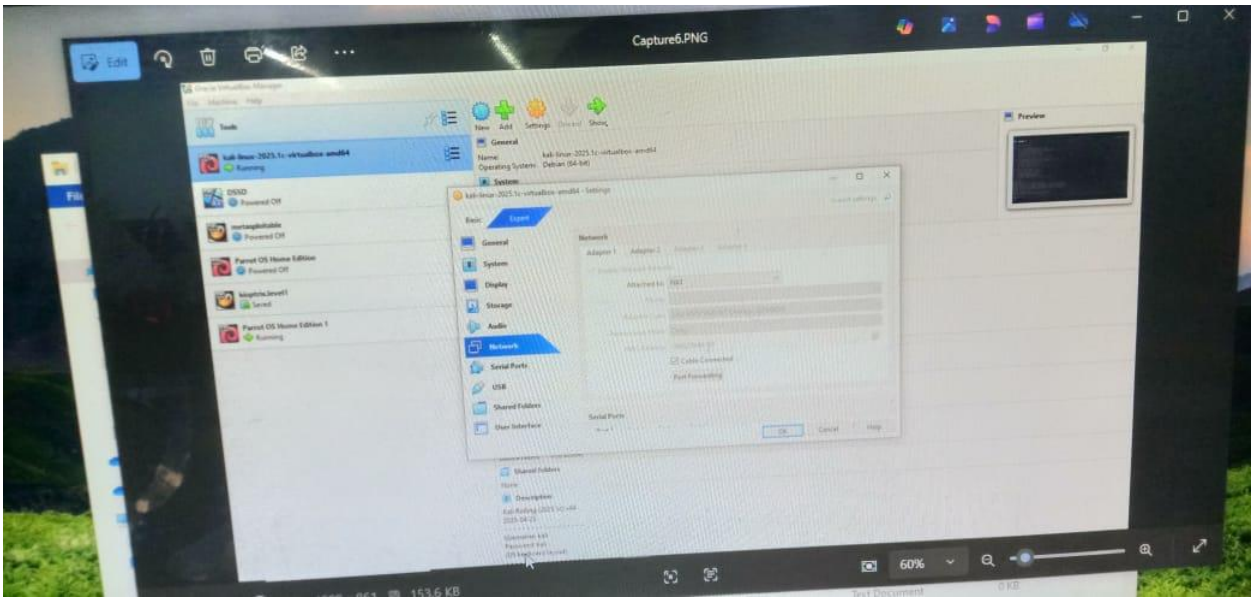
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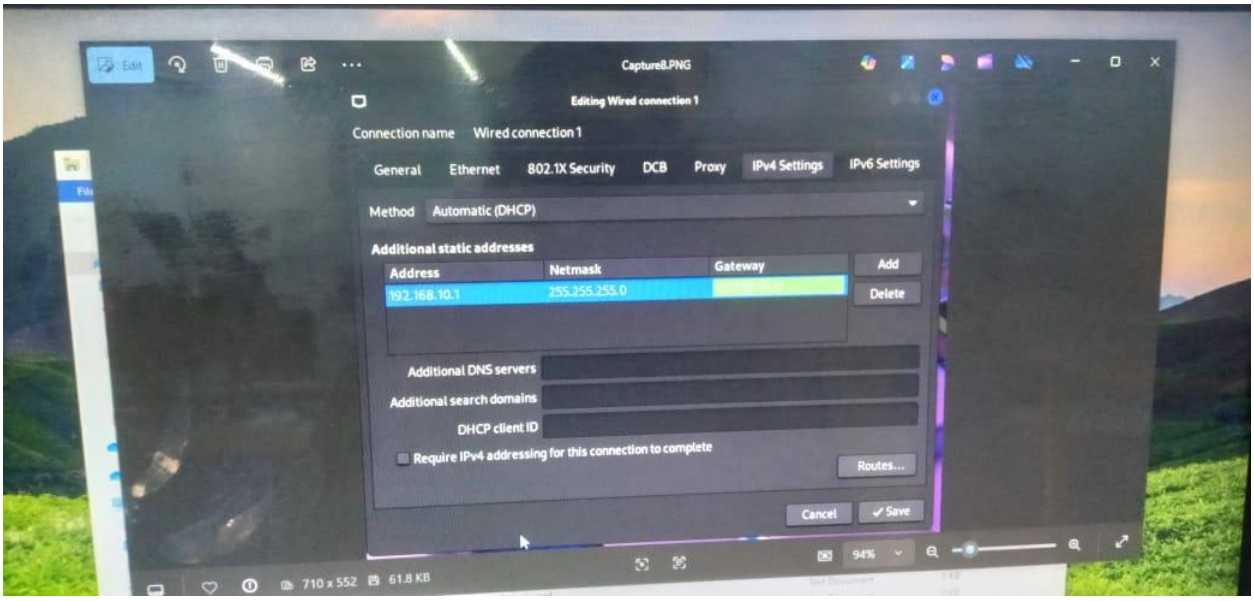
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Address
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Gateway
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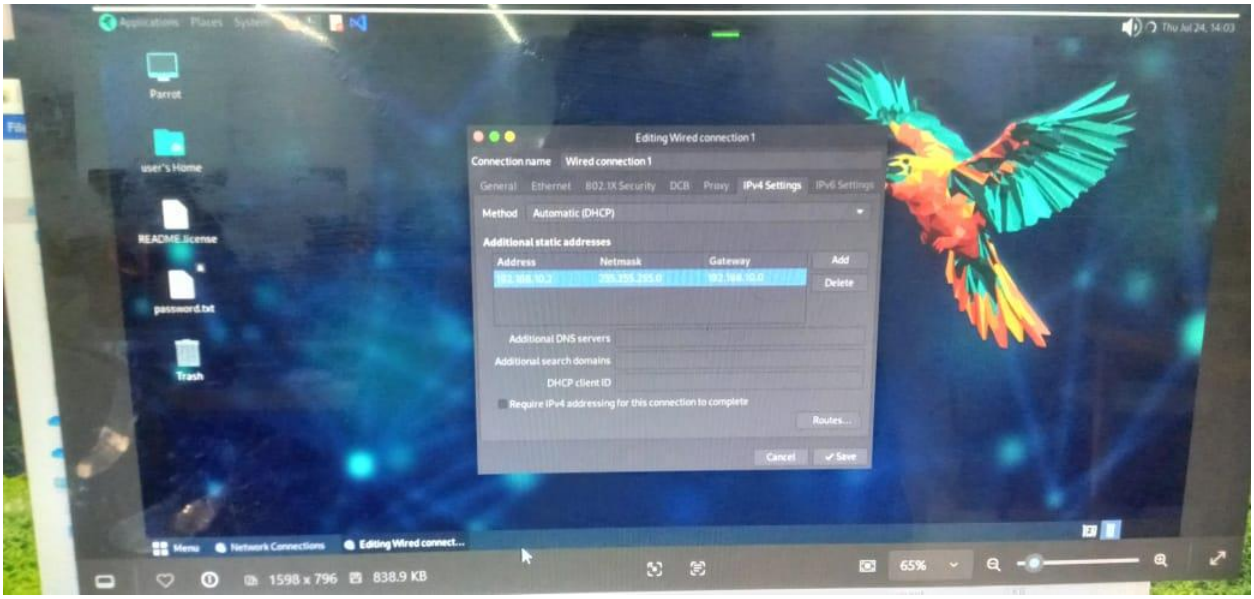
Parrot
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802.1X Security
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Proxy
IPv4 Settings
IPv6 Settings
Method
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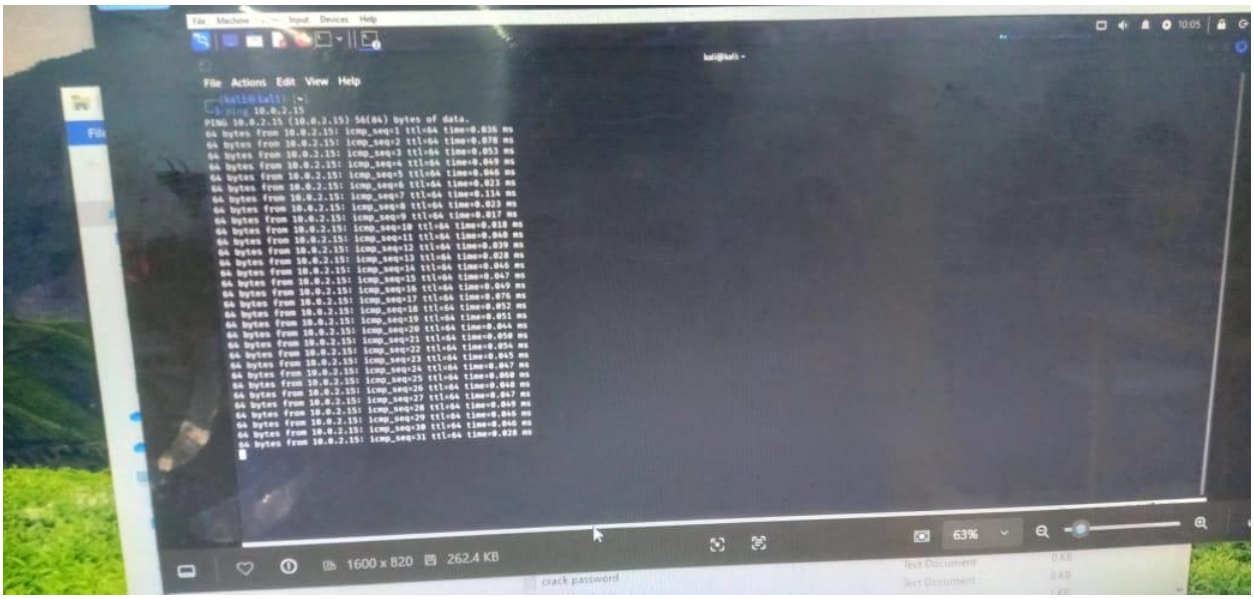
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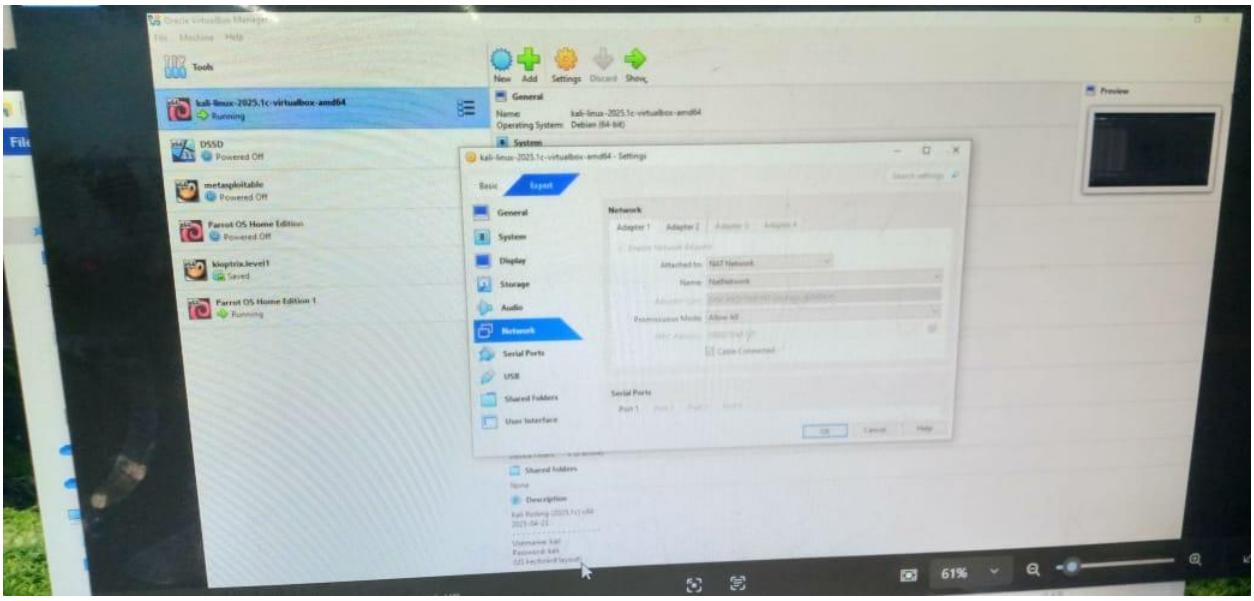
Tools
kali-linux-2025.1c-virtualbox-amd64
DSSD
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Parrot OS Home Edition
kioptrix.level1
Parrot OS Home Edition 1
General
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Serial Ports
USB
Shared Folders
User Interface
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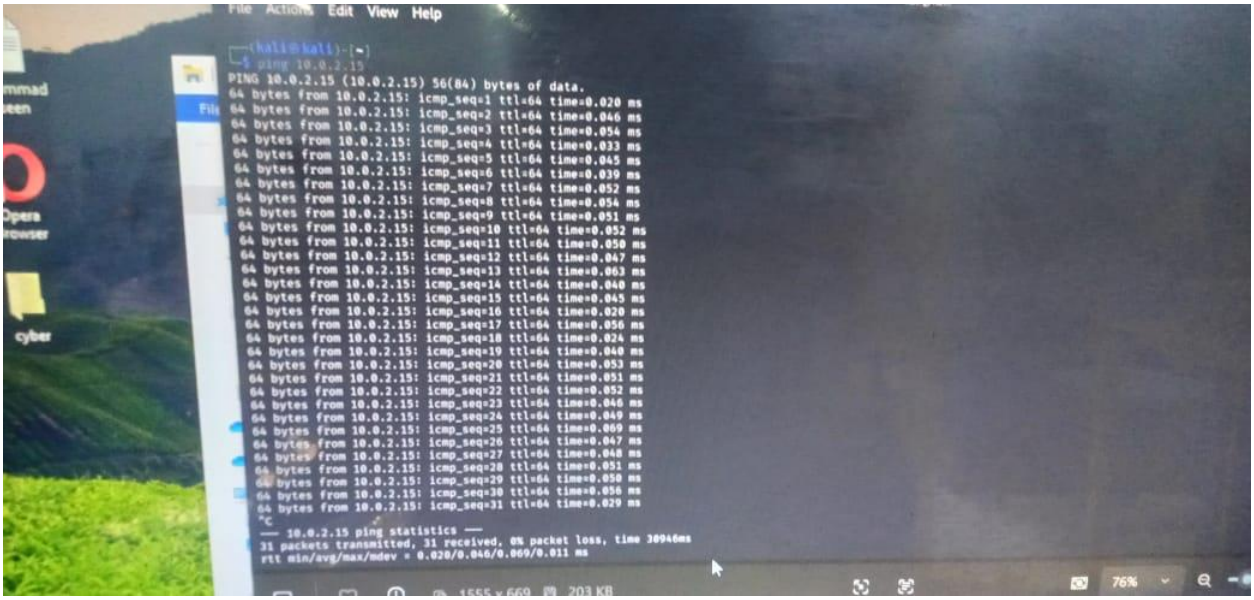


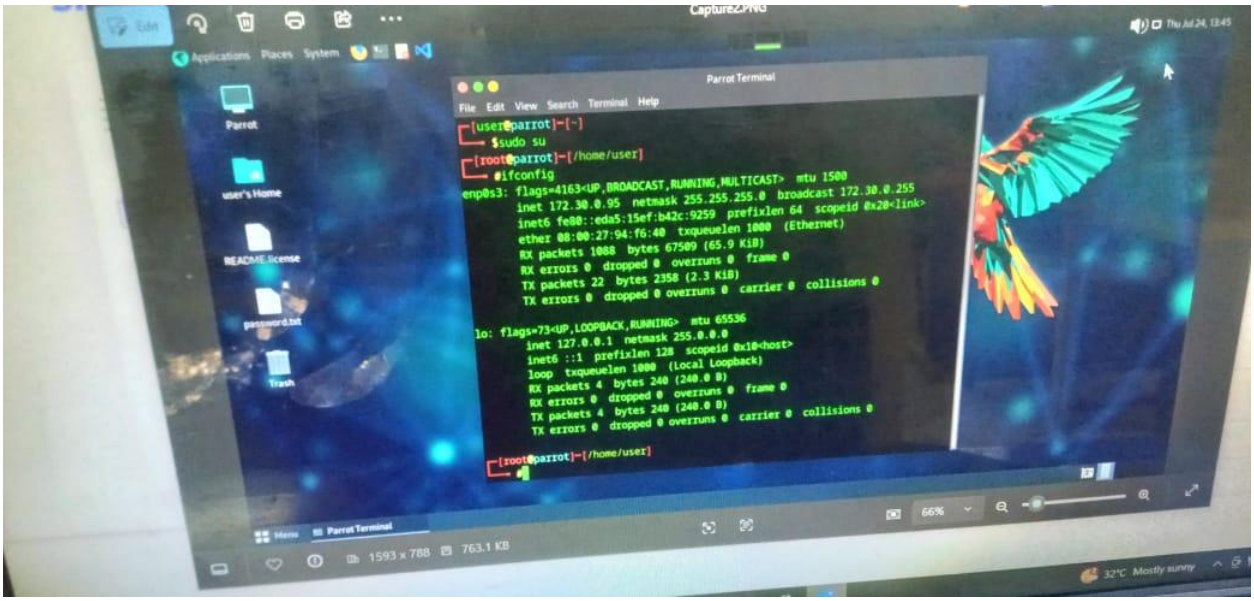


## Learning Outcomes

Through this activity, I learned how to set up and configure different VirtualBox network modes such as NAT, Bridged, and Internal Network. I understood how each mode works and how it affects communication between virtual machines and external networks. I also practiced checking IP addresses and testing network connectivity using basic commands like ping,ifconfig and ip a Additionally, I learned how to identify and fix common networking issues inside virtual machines. Overall, this gave me practical experience in connecting two different operating systems and managing their communication in a virtual lab environment.

## Conclusion

By completing this activity, I understood how different VirtualBox network modes work and how they are used in real-life network setups. This helped me build a strong foundation in virtual networking and system communication. It also prepared me for future tasks like simulating attacks, creating isolated environments, and learning advanced cybersecurity skills.



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